MYSQL ASSIGNMENT 2

```
USE employee;


-- 1. DISTINCT VALUES
-- Retrieve distinct salaries
SELECT DISTINCT salary
FROM Employees;



-- 2. ALIAS (AS)
-- Rename age and salary columns
SELECT
   age AS Employee_Age,
   salary AS Employee_Salary
FROM Employees;



-- 3. WHERE CLAUSE AND OPERATORS


-- A. Employees earning more than 50000 and hired before 2016-01-01
SELECT * FROM Employees
WHERE salary > 50000
  AND hire_date < '2016-01-01';



-- B. Find employees with missing designation
```

```
SELECT * FROM Employees
WHERE designation IS NULL
  OR TRIM(designation) = '';

-- Fill missing designations
UPDATE Employees
SET designation = 'Data Scientist'
WHERE designation IS NULL
  OR TRIM(designation) = '';



-- SORTING AND GROUPING DATA
-- ==========================================

-- 4. ORDER BY  Department ascending, salary descending
SELECT * FROM Employees
ORDER BY department_id ASC, salary DESC;



-- 5. LIMIT   First 5 employees hired in 2018
SELECT * FROM Employees
WHERE hire_date >= '2018-01-01'
 AND hire_date < '2019-01-01'
ORDER BY hire_date ASC, employee_id ASC
LIMIT 5;



-- 6. AGGREGATE FUNCTIONS
```

```
-- A. Total salary in Finance department
SELECT SUM(e.salary) AS Total_Finance_Salary
FROM Employees AS e
INNER JOIN Departments AS d
   ON e.department_id = d.department_id
WHERE d.department_name = 'Finance';



-- B. Minimum age
SELECT MIN(age) AS Minimum_Age
FROM Employees;



-- 7. GROUP BY

-- A. Maximum salary for each location
SELECT l.location_name,MAX(e.salary) AS Maximum_Salary
FROM Location AS l LEFT JOIN Employees AS e
   ON l.location_id = e.location_id
GROUP BY l.location_id, l.location_name;



-- B. Average salary for each designation containing the word Analyst
SELECT designation, AVG(salary) AS Average_Salary FROM Employees
WHERE designation LIKE '%Analyst%' GROUP BY designation;
```

```
-- 8. HAVING

-- A. Departments with fewer than 3 employees
SELECT
   d.department_id,
   d.department_name,
   COUNT(e.employee_id) AS Total_Employees
FROM Departments AS d
LEFT JOIN Employees AS e
   ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
HAVING COUNT(e.employee_id) < 3;



-- B. Locations where female employees
-- have an average age below 30
SELECT
   l.location_name,
   AVG(e.age) AS Average_Age
FROM Location AS l
INNER JOIN Employees AS e
   ON l.location_id = e.location_id
WHERE e.gender = 'Female'
GROUP BY l.location_id, l.location_name
HAVING AVG(e.age) < 30;
```

```
-- JOINS
-- ==========================================

-- 9. INNER JOIN
-- Employee names, designations, department names
SELECT
   e.employee_name,
   e.designation,
   d.department_name
FROM Employees AS e
INNER JOIN Departments AS d
   ON e.department_id = d.department_id;


-- 10. LEFT JOIN
-- All departments and employee counts
SELECT
   d.department_id,
   d.department_name,
   COUNT(e.employee_id) AS Total_Employees
FROM Departments AS d
LEFT JOIN Employees AS e
   ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
ORDER BY d.department_id;


-- 11. RIGHT JOIN
```

```
-- All locations and assigned employee names
SELECT
   l.location_id,
   l.location_name,
   e.employee_name
FROM Employees AS e
RIGHT JOIN Location AS l
   ON e.location_id = l.location_id
ORDER BY l.location_id, e.employee_name;
```